

Can graph theory analysis of brain networks characterize differences in connectivity between brain regions in Alzheimer's patients versus control?

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**Honors Thesis
Arizona State University**

May 2026

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative condition, necessitating early diagnosis and intervention. This study investigates whether graph theory analysis of resting-state functional MRI (rs-fMRI) can characterize differences in brain network connectivity between AD patients and cognitively normal (CN) controls. Neuroimaging data for 20 AD, and 20 CN subjects was selected from the Alzheimer's Disease Neurological Initiative (ADNI) database using the ADNI-2 cohort. After preprocessing, functional connectivity matrices were constructed using the RM96 brain parcellation with a 30% sparsity threshold. Then, local and global graph theory measures were computed. Group differences between CN and AD groups were analyzed using Cohen's d tests for each region of interest (ROI) and visualized on a functional connectome. The largest differences were primarily in the frontal and medial temporal regions. An ML model using the Support Vector Machine algorithm was developed for the purpose of AD prediction. This study uses key graph theory measures for a comprehensive understanding of brain network topology and functional connectivity in AD. Through hyperparameter tuning and principal component analysis, this model demonstrates moderate predictive capabilities with an accuracy of 75%. This shows that graph theory measures derived from rs-fMRI data detect moderate differences between AD and CN subjects.

Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder, and an estimated 7.2 million Americans aged 65 and older live with Alzheimer's dementia today. This number could grow to 13.8 million by 2060, barring the development of medical breakthroughs to prevent or cure AD ("2025 Alzheimer's Disease Facts and Figures," 2025). Alzheimer's disease is caused by an accumulation of Amyloid β and neurofibrillary tangles in the hippocampus and other brain regions (Kamatham et al., 2024). An accumulation of neurofibrillary tangles block connections from the synapse intracellularly, eventually causing neuronal death. In contrast, Amyloid β plaques form extracellularly around the neuron, also causing neuronal death. Neuronal death leads to a loss of connection between neural circuits and brain regions. This causes progressive cognitive decline in the form of short-term memory loss, decreased motor control, and loss of speech (aphasia). Despite decades of research on this disease, no cure has been developed. This is partially due to clinical diagnosis of Alzheimer's disease occurring when substantial neurodegeneration has already occurred in the patient. This highlights the need for biomarkers that can accurately detect and characterize Alzheimer's disease, ideally in the early stages of the disease before irreversible damage to the brain occurs.

Resting-state functional magnetic resonance imaging (rs-fMRI) has emerged as a promising neuroimaging tool for identifying biomarkers and for staging of AD. To acquire rs-fMRI data, the Blood Oxygen Level Dependent (BOLD) signal is measured, which captures

fluctuations in neural activity across brain regions. The brain is in a resting state and does not participate in tasks during this acquisition. The statistical relationships are then measured between BOLD time series extracted from different brain regions. Functional connectivity matrices that show correlations between the BOLD time series activity of any two brain regions are created. Disruptions to these functional connectivity patterns have been consistently observed in Alzheimer's disease, particularly in frontal and temporal regions and within the default mode network (Gu et al., 2020; Zhao et al., 2017; Shi et al., 2020). The ADNI (Alzheimer's Disease Neuroimaging Initiative) database shares recorded rs-fMRI data to be used for public use. This data is used for this study.

The application of graph theory to the analysis of brain data allows the brain to be considered as a complex system of nodes (brain regions) and edges (functional connectivity between nodes). Graph theory measures can be calculated based on the relationships between nodes, edges, and between large partitions of the brain network. The values of graph theory measures can provide insight on the architecture of the brain, key connectivity hubs, and disease related structural or functional changes.

Prior work has established that Alzheimer's disease disrupts the balance between functional segregation and integration in the brain, where healthy brains exhibit modular organization. There are two keys to modular organization. First, clusters of densely connected regions (modules) that support specialized functions are needed. Efficient, minimal connections between modules are also used in the brain to ensure efficient communication across the brain (Barabino et al., 2024; Betzel et al., 2017). In AD, this balance breaks down: local clustering tends to decrease, reflecting a loss of specialized local processing, while global communication efficiency is also impaired. These disruptions have been observed consistently across studies using ADNI data and a range of graph theory metrics, including average strength, clustering coefficient, local efficiency, modularity, and transitivity (Tijms et al., 2013). Graph theory features derived from rs-fMRI have also been used successfully as inputs to machine learning models. For example, Karim et al. (2024) derived graph theory measures from functional connectivity networks and tested various machine learning classifiers for AD prediction across two publicly available datasets. Through 5-fold cross-validation, all models demonstrated substantial predictive capabilities with accuracies ranging from 82-92% with a support vector machine achieving the best performance. This study identified discriminative features for understanding brain network topology and functional connectivity in AD, suggesting the translational potential of integrating machine learning, graph theory, and rs-fMRI analysis for AD prediction.

The goal of this study is to use network analysis of brain networks (through graph theory) to characterize differences in connectivity between brain regions in Alzheimer's patients versus control (CN). First, a processing pipeline is used with a RM96 parcellation to parcellate brain regions and establish functional connectivity between brain regions. Then, a range of global and local graph theory measures are computed. Group differences between the values of the measures in the AD and CN group were found for the various graph theory measures. Finally,

the original values of the graph theory measures are used as features in a Support Vector Machine (SVM) classifier to distinguish AD from control subjects. Overall, this analysis aims to show that machine learning can be used to classify Alzheimer's patients in a computationally efficient way.

Methods

This study is a cross-sectional analysis using publicly available data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (ADNI, 2026). Resting-state fMRI data from 20 participants diagnosed with Alzheimer's disease (AD), along with 20 healthy controls (CN), were obtained through the ADNI-2 cohort. Participants in this study were recruited between 2011 and 2013 through the ADNI-2 protocol. Participants were chosen with an age range of 70-90 years old, and those selected had completed both 3D MPRAGE and resting-state fMRI data.

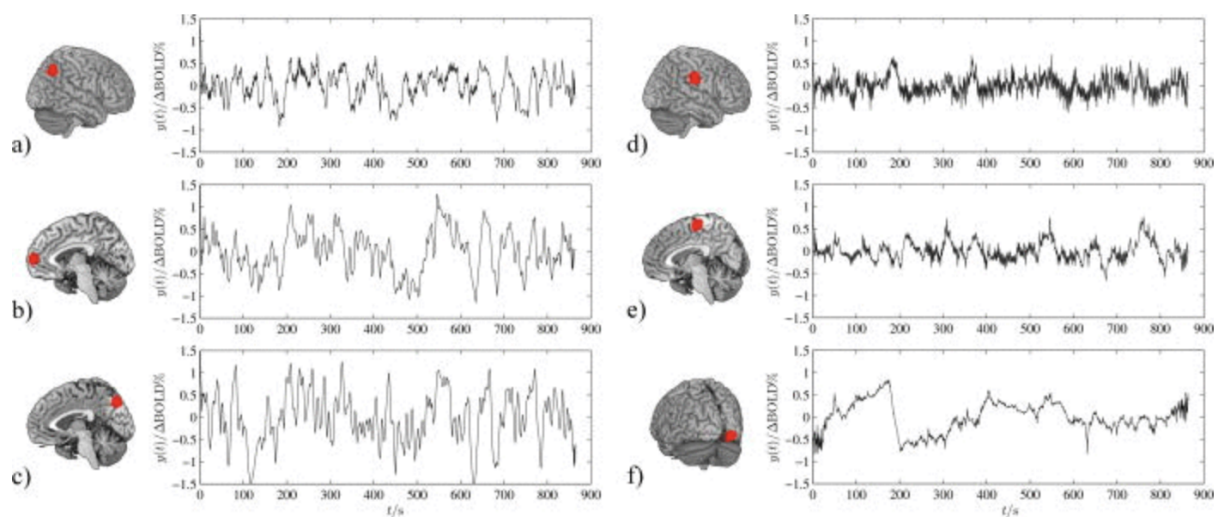


Figure 1: Examples of BOLD time series. fMRI detects the Blood Oxygen Level Dependent (BOLD) signal across the brain. This example depicts the BOLD signal associated with six different brain regions, where panels a-f show the BOLD time series signal over a period of 900 seconds. The vertical axis depicts the BOLD signal value, and the horizontal axis represents time in seconds (from Minati et al., 2015).

Imaging Protocol

Functional Magnetic Resonance Imaging (fMRI) data were obtained from a Philips Achieva 3.0-Tesla Scanner with repetition time (TR) of 3,000 ms and echo time (TE) of 30 ms,

and 140 temporal volumes were acquired for each participant in a single scanning session. More specific information can be obtained from the ADNI website. T1-weighted 3D MPRAGE scans were acquired for the same participants on a Philips Achieva 3.0-Tesla scanner (Philips Medical Systems) using a SENSE-Head-8 coil. Acquisition parameters were: sagittal plane; 3D gradient-echo sequence; TR = 6.8 ms; TE = 3.1 ms; flip angle = 9° ; matrix = $256 \times 256 \times 170$; voxel size = $1.0 \times 1.0 \times 1.2$ mm; slice thickness = 1.2 mm. Functional images were collected on the same scanner using a gradient-echo sequence with TR = 3000 ms, TE = 30 ms, flip angle = 80° , matrix = 64×64 , voxel size = $3.3 \times 3.3 \times 3.3$ mm, slice thickness = 3.3 mm, and 140 temporal volumes per session.

During fMRI studies, the Blood Oxygen Level Dependent (BOLD) signal is detected across the brain. The BOLD time series signal represents changes in blood oxygenation which have been shown to be correlated with neural activity. When neural activity increases, there are changes in local blood flow and oxygenation, which alters the balance of oxygenated and deoxygenated hemoglobin. This produces the BOLD signal, which is detected by fMRI. Examples of BOLD time series are shown in Figure 1.

Data Preprocessing

The CONN package for MATLAB is an open-source cross-platform software for the computation, analysis, and visualization of functional connectivity. The CONN default preprocessing pipeline was used in this study (Whitfield-Gabrieli & Nieto-Castanon, 2012). This pipeline includes steps for functional realignment, slice-timing correction, outlier identification, direct segmentation and normalization, and functional smoothing. Determining the functional connectivity requires the use of a brain parcellation (also called a brain atlas), which separates the human brain into distinct areas according to their structure or function (Eickhoff et al., 2018). Many different types of brain parcellations exist depending on the needs of the user. In this study, the Regional-Map 96 (RM96) parcellation was used, which is composed of 82 cortical sections (regions of interest or ROIs) and 14 subcortical ROIs (Bezgin et al., 2017). CONN registers (aligns) the desired parcellation onto the T1-MRI image provided by the user so that the T1-MRI image is labeled according to the ROI specifications of the parcellation.

For each ROI, BOLD signals are averaged to obtain a time series associated with that ROI. Then a Pearson correlation is calculated between every pair of time series to give the functional connectivity value for each possible pair of ROIs. The Pearson correlation values then are normalized using a Fisher-z transformation to ensure that the data are normally distributed (Salkind, 2007). The result is a functional connectivity matrix with rows and columns that are associated with the ROIs, where each element of the matrix represents the functional correlation of connectivity between the ROIs associated with row i and column j (Whitfield-Gabrieli & Nieto-Castanon, 2012).

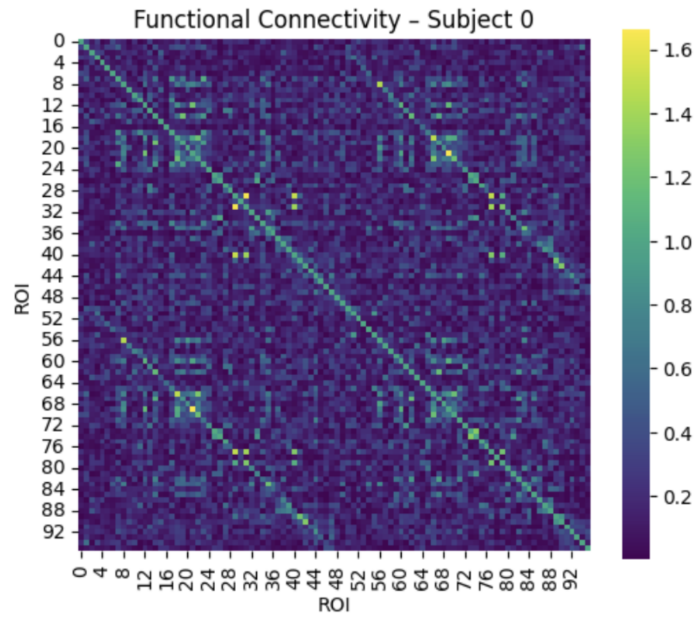


Figure 2: Functional connectivity matrix for one subject. Each row (and column) represents one of the 96 regions of interest (ROIs) that make up the brain parcellation. Matrix cell values are represented by color and depict the strength of the correlation of firing activity between the ROIs associated with that row and column.

Specifically, each cell's functional connectivity is computed using

$$R_{ij} = z_{ij}(BOLD_i, BOLD_j) \quad (1.1)$$

where R_{ij} represents the functional connectivity value for a cell in row i and column j , and z_{ij} represents the Pearson correlation. This is calculated as

$$z_{ij} = \frac{1}{2} \ln \left(\frac{1+r_{ij}}{1-r_{ij}} \right). \quad (1.2)$$

In this equation, z represents the Fisher-z transformed correlation coefficient, and r represents the Pearson correlation coefficient between two BOLD time series. Pearson correlation is commutative, meaning that the correlation between regions i and j is equal to the correlation between regions j and i , which results in a symmetric matrix. On the diagonals of the functional connectivity matrix, same element pairs are computed. This results in a value of 1 being computed for each element on the diagonal of the functional connectivity matrix.

Graph theory representation

We can associate each ROI in the brain parcellation with a node (or vertex) of a mathematical structure known as a graph, and each functional connection (correlation) between two regions is an edge in the graph. Each edge is assigned a weight that reflects the associated correlation value

in the connectivity matrix. In this way, each functional connectivity matrix is associated with a corresponding graph. Since these matrices are symmetric, the graphs associated with functional connectivity matrices are undirected.

Since a connectivity matrix typically has non-zero values in each cell, an associated graph would be fully connected. Thus, connectivity matrices typically are processed further using a thresholding criterion that removes the weakest connections that are not associated with physiological connections. In this analysis, a sparsity threshold of 30% was used, which removes the weakest 30% of connections by replacing the low-value entries in the functional connectivity matrix with zeroes. Additionally, for this analysis, the absolute value of each entry of the connectivity matrix is used. This approach assumes that both positive and negative correlations are associated with connectivity, and further analysis only assumes connectivity, ignoring the sign of the connection. This approach simplifies graph characterization since many graph measures cannot account for negative weights for edges.

After this pre-processing, a graph is associated with each individual's rs-fMRI connectivity matrix, and for further analysis, the graph is an undirected graph with normalized, positive weights.

Graph theory metrics

Once neuroimaging data has been processed to produce connectivity matrices and associated graphs, graph theory analysis is used to characterize the connectivity properties of these graphs. Both local and global measures are explored in this analysis. Local measures represent information about a specific brain region and its relationships with other brain regions. Global measures represent metrics concerning the entire brain rather than individual regions.

Metrics	Definition
Node Strength (L)	Sum of values of edges connected to a node
Local Clustering Coefficient (L)	Describes how interconnected neighboring nodes are to each other
Characteristic Path Length (G)	The average shortest path length in a network
Efficiency (G)	The average inverse shortest path length in the graph. Note: this is not equivalent to the inverse of characteristic path length.
Small-world Coefficient(G)	A network has small world properties if it displays high local clustering (segregation)

	and short characteristic path length (integration)
Modularity (G)	Density of edges inside communities compared to edges between communities.
Local Efficiency (L)	The inverse shortest path length of a node
Betweenness Centrality (L)	The fraction of times a node is involved in the shortest path between any two other nodes out of all total shortest paths between two nodes.
Participation Coefficient (L)	How evenly a nodes connections spread across a network

Table 1: Summarized definitions for graph analysis measures calculated on the connectivity matrix (Rubinov & Sporns 2010; Zimmermann et al. 2016; Newman 2006; Muldoon, Bridgeford, and Bassett 2016). L designates a local measure, and G designates a global measure.

Strength: Undirected strength is a local measure defined as the sum of the weights of all edges that are linked to a node (Fornito, 2016).

Local Clustering Coefficient: The local clustering coefficient is a measure of functional segregation, which describes how interconnected neighboring nodes are to each other. It measures the density of connections (triangles) within a node's neighborhood and factors in the weights of those connections. This is quantified by

$$C_i = \frac{2}{k_i(k_i-1)} \sum_{j,k} (w_{ij}w_{jk}w_{ki})^{1/3}, \quad (1.3)$$

where C_i represents the local clustering coefficient value for node i . w_{ij} represents the connection strength between node i and a neighboring node j . The value k_i represents the number of edges connected to node i , which is called the degree of i . Note that the expression $k_i(k_i-1)/2$ describes the number of possible edges among the neighbors of node i . For each possible pair of neighbors of node i , if such nodes j and k are connected, these nodes form a triangle with node i . In this case, the node weights from the edges in that triangle are multiplied, and then the geometric mean is calculated. This quantity is then summed over all the possible pairs of neighbors of node i . If j and k are not connected by an edge, then the product of the weights is 0. This quantity is then normalized once more between 0 and 1 and characterizes the tendency of the neighbors of node i to be interconnected with each other (Onnela et al., 2005).

Characteristic Path Length: The characteristic path length is a global feature and is defined as the average of the shortest path lengths computed across all nodes in the network. This can be computed by turning the distance matrix into a vector by stacking each column sequentially and computing the mean of the vector (Hughes, 2025).

Global Efficiency: Efficiency is the average inverse shortest path length of the graph (Hughes, 2025). Global Efficiency is known as a reliable measure of integration for a network.

$$E_{glob}(G) = \frac{1}{N(N-1)} \sum_{i \neq j \in G} \frac{1}{d_{ij}} \quad (1.4)$$

Here, G represents the graph under consideration, N represents the number of nodes in the graph, and d represents the weighted shortest path length between nodes i and j . Note that the inverse of the shortest path length is calculated for every pair of nodes i and j in G . Then these values are averaged using the number of all possible connections, $N(N-1)$.

Local Efficiency: Local efficiency reveals the robustness of the network and shows how efficient communication is between the neighbors of i when i is removed (Rubinov & Sporns, 2010).

$$E_{loc}^w = \frac{1}{2} \sum_{i \in N} \frac{\sum_{j,h \in N, j \neq i} (w_{ij} w_{ih} (d_{jh}^w(N_i)^{-1})^{1/3}}{k_i(k_i-1)} \quad (1.5)$$

Here, N represents the set of all nodes, k_i represents the degree of node i , and w_{ij} represents the weight of the connection between nodes i and j . $d_{jh}(N_i)$ is the length of the shortest path between j and h that contains only neighbors of i . Taking the inverse of this expression transforms the expression into a measure for efficiency, rather than path length. A shorter path length contributes strongly to high local efficiency, and a longer path length contributes strongly to low local efficiency. Furthermore, the expression has an exponent of $1/3$ to balance the influence that connection weights and the inverse path lengths have on the overall expression. This is summed for all nodes that are neighbors of node i , and then normalized by $k_i(k_i-1)$ which represents the maximum possible number of connections among the nodes that are neighbors of node i . Finally, the calculation sums this expression for all nodes i in the graph, and divides by 2 to ensure that connections between node i and its neighbors are not double counted.

Small-world Coefficient: Small-world graphs display high local clustering with short average path lengths. This global graph theory measure describes to what extent a network has a ratio of high local clustering (segregation) and short characteristic path length (integration) (Bassett, 2016). This is described using the formula

$$\sigma_{weighted} = \frac{\Gamma_{weighted}}{\Lambda_{weighted}} \text{ where } \Lambda = \frac{L_{brain}}{L_{random}}, \Gamma = \frac{C_{brain}}{C_{random}} \quad (1.6)$$

The small-world coefficient relies on three parameters. Γ (segregation) represents to what degree a network is closely clustered compared to a randomized network with the same number of

nodes and edges. Λ (integration) represents how efficiently information travels between distant nodes. σ combines segregation and integration into a single measure by describing their ratio.

First, the weighted clustering coefficient, cc , and the mean clustering coefficient, C , are computed by averaging the local clustering coefficients across all nodes. Similarly, a randomized network with preserved degree distribution and connectivity is generated in order to calculate the randomized clustering coefficient and its mean, respectively. Weights of nodes are converted into original and randomized lengths. Those lengths are used to calculate a distance from an original and randomized matrix in the following step. This intermediate step is necessary to then compute the original and randomized characteristic path lengths. Finally, the three parameters for the small world coefficient are computed.

Modularity: Modularity is a global measure that optimizes the within group connections and minimizes out-of-group connections. Newman's spectral community detection method was used to identify the optimal community structure of each brain graph (Newman, 2006), which is contained in the *modularity_und.m* function in the BCT (Rubinov & Sporns, 2010). This method is an iterative, deterministic method which aims to maximize the modularity metric, Q . The modularity Q for a given division of a network is defined to be the fraction of edges within groups minus the expected fraction of such edges in a randomized null model of the network (Newman, 2013). Q is defined as

$$Q = \frac{1}{2m} \sum_{ij} [A_{ij} - P_{ij}] \delta(g_i, g_j) . \quad (1.7)$$

The value of δ is 1 if the nodes belong to the same group and 0 if they do not. A_{ij} is the original adjacency (connectivity) matrix value, and P_{ij} is the expected connectivity value for the same connection. $1/2m$ is included for compatibility with other definitions of modularity, where m is the total number of edges, and ensures that node pairs are not counted twice. Due to underlying assumptions about the null model that the expected connectivity value is being pulled from, the probability of the connection matrix can be explicitly defined as $P_{ij} = k_i k_j$, where k_i is the degree of node i (Newman, 2006). The Newman spectral method creates a modularity matrix, B , where $B_{ij} = A_{ij} - P_{ij}$. Then groups are identified by finding the eigenvector that corresponds to the most positive eigenvalue of the modularity matrix and assigning groups based on the sign of the elements in the associated eigenvector. This initial community structure assignment is improved by switching individual node assignments once - if the switch improves modularity this is kept as the community assignment (Rubinov & Sporns, 2010). This initially splits the graph into two communities but will then further split those two communities into four communities (two from each) and so on. A community is only further split if it improves the overall modularity value.

Node Betweenness Centrality: Node betweenness centrality is a graph theory measure that represents the fraction of all shortest paths in the network that contain a given node. Nodes with high values of betweenness centrality participate in a large number of shortest paths (Rubinov & Sporns, 2010). That is, for every pair of nodes in a connectivity matrix, there exists at least one

shortest path between the vertices. That means that at least one path exists such that the sum of the weights of the edges (for weighted graphs) is minimized. The formula for betweenness centrality can be defined as

$$b_i = \frac{1}{(n-1)(n-2)} \sum_{h,j \in N, h \neq j, h \neq i} \frac{\rho_{hj}(i)}{\rho_{hj}} \quad (1.8)$$

where the value ρ_{hj} is the number of shortest paths between h and j , and $\rho_{hj}(i)$ is the number of shortest paths between h and j that pass-through i .

Participation Coefficient: The participation coefficient assesses the diversity of intermodular interconnections of nodes and represents the participation of node i in the modular structure of the network. Nodes with a high within-module degree but with a low participation coefficient (known as provincial hubs) are likely to play an important part in the facilitation of modular segregation. On the other hand, nodes with a high participation coefficient (known as connector hubs) are likely to facilitate global intermodular integration (Rubinov & Sporns, 2010). The participation coefficient is calculated as

$$y_i^w = 1 - \sum_{m \in M} \left(\frac{k_i^w(m)}{k_i^w} \right)^2. \quad (1.9)$$

Here, M is the set of modules (see modularity), and $k_i(m)$ is the sum of weights from node i to the nodes in module m . Each term in the summation normalizes the proportion of node i 's edges that connect with module m by dividing by the sum of the weights of all edges. Squaring this expression allows this quantity to have more influence in the resulting value of the participation coefficient (because it is subtracted from 1). A low value suggests that a node's connections are confined to its own module.

Other Analysis and Visualization

Cohen's d Test (Node Level Analysis): Cohen's d is a statistical test to measure the difference in means between independent groups (Cohen, 2009). Figure 4 depicts the difference between AD and CN for each of four local measures, where values are shown for the ROIs with the largest Cohen's d values. The stronger the contrast, the more of a role the included ROI plays in Alzheimer's disease.

In some cases, we visualize results using plots which represent results spatially relative to brain structure (Bullmore & Sporns, 2009). In this analysis, the python library *nilearn* is used to visualize the brain's structure. We employ *nilearn.plotting.plot_connectome*, which uses three-dimensional neuroimaging data to create projected two-dimensional visual representations of the brain (Pedregosa et al., 2011). An RM96 atlas is used to match the parcellation for visualization to that used in analysis. This atlas contains ROI names and MNI (Montreal Neurological Institute) coordinates mapped to (x,y,z) in *mm*. The MNI coordinates together with a matrix are used to visualize the nodes associated with this parcellation. The lateral, medial, and axial brain

model views depict the location of the greatest standardized group differences between Alzheimer's disease and control individuals for the local graph theory metrics. In addition to the functional connectome, a table is created to provide precise information about all the nodes in different sectors of the brain.

Machine Learning: A nonlinear SVM (Support Vector Machine) was used on feature level matrices to develop a classifier that can be used to predict whether a subject has Alzheimer's disease. An SVM is a machine learning model designed for two group classification problems. A linear decision boundary is created in the feature space that divides the two classified groups (Cortes & Vapnik, 1995). The feature space is the space defined by the axes where subjects are given a value based on the amount of variance the subject holds in relation to the dataset. Both global and local features were combined to capture the full scope of the dataset. Before model development, all features were cleaned of any non-numerical symbols or columns with constant measures. Hyperparameters were tuned using a grid search with stratified 5-fold cross-validation. The tuned parameter grid included linear and radial basis function kernels, several values of C, and both automatic and scale-based gamma settings. The model was evaluated using the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) score. The ROC curve plots the subjects that are correctly classified (true positive rate) against the subjects that are incorrectly classified (false positive rate) using the SVM model. The curve is plotted on all possible classification thresholds from 0 to 1.0. The AUC score summarizes the curve as a single value describing the model's ability to distinguish subjects in the AD and CN group.

Principal Component Analysis: Principal Component analysis is a statistical tool used in data analysis to reduce the number of dimensions, while retaining as much possible of the data's variation. The data is split into the first few principal components containing most of the data's variation. The visualization and statistical analysis of these new variables can help find similarities and differences between samples (Groth et al., 2013). In this study a two-dimensional PCA projection was constructed by projecting each subject onto the first two principal components. Subjects are labeled according to their diagnosis (CN or AD). The decision boundary that the SVM model constructs separates the data points onto two regions. One region corresponds to the subjects that the model predicted have Alzheimer's disease while the other regions correspond to the subjects that the model predicted do not have Alzheimer's disease. This allows visual assessment of group overlap and separation in the feature space. This visualization only captures a subset of the total variance and does not represent the full dimensional feature space used by the nonlinear SVM model.

Results

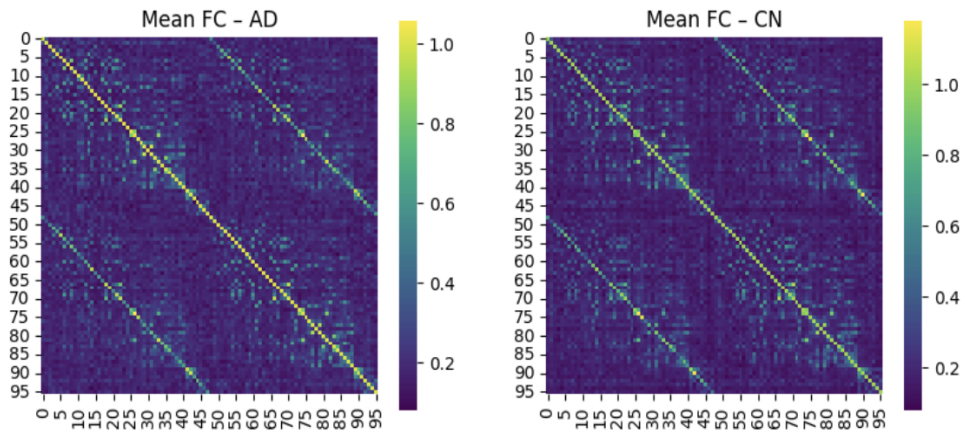


Figure 3: Averaged functional connectivity matrices. The average connectivity matrix associated with Alzheimer’s disease (AD) patients is shown on the left, and the average connectivity matrix for cognitively normal (CN) subjects is shown on the right. The rows and columns represent all 96 regions of interest, and each cell entry in a matrix represents the functional connectivity value for the regions associated with that row and column.

The modular structure that is seen in the average functional connectivity matrices in Figure 3 shows that brain regions form block-like clusters with increased connectivity among regions that share similar function, while connections to dissimilar regions are sparse and weakly connected.

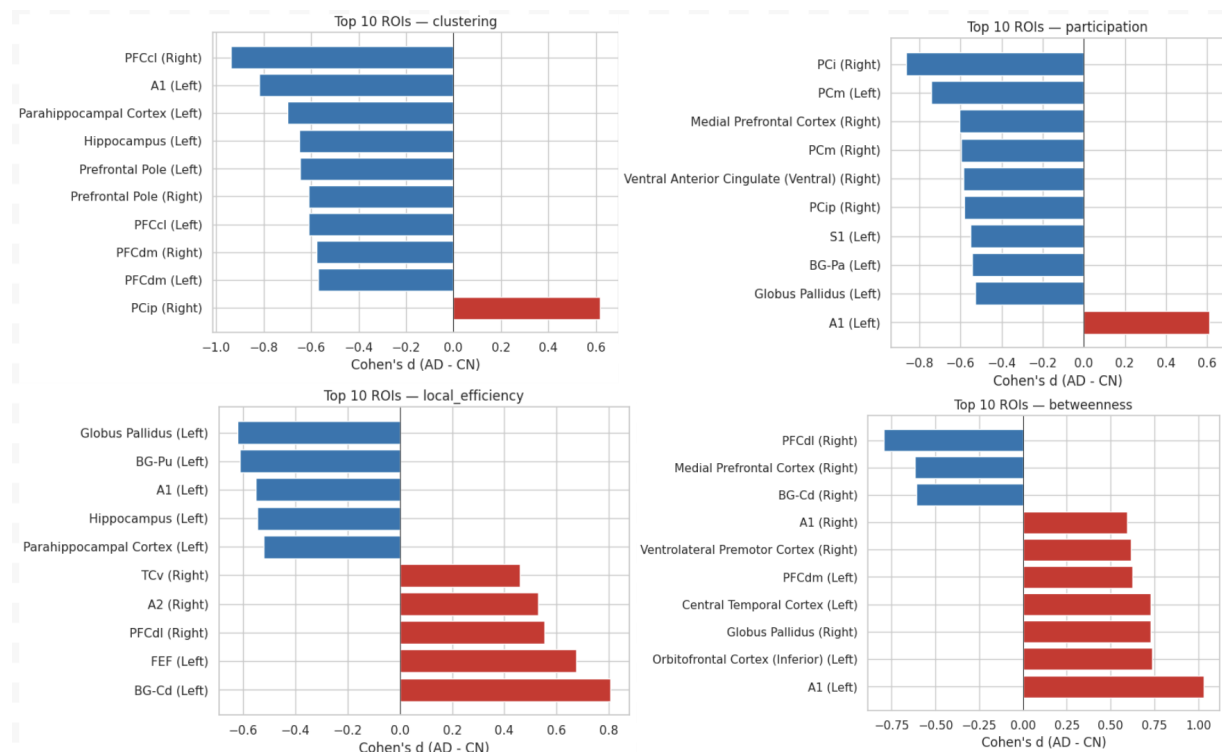


Figure 4a: ROIs with largest standardized mean differences between AD and CN. Each panel shows the 10 ROIs with the largest Cohen's d values for a specific graph theory measure. Group differences are visualized using horizontal bars. Red bars indicate positive standardized mean differences (AD value is larger), and blue bars indicate negative standardized mean differences (CN is larger). Graph theory measures include local clustering coefficient (upper left), participation coefficient (upper right), local efficiency (lower left), and betweenness centrality (lower right)

Figure 4 summarizes the results of Cohen's d calculations based on the average connectivity depicted in Figure 3. As we can see from Cohen's d values for the local clustering coefficient, most of the top ROIs are blue, indicating negative effect sizes (AD < CN). This suggests reduced connectivity in these regions in Alzheimer's disease patients compared to the brains of cognitively normal controls. Reduced local clustering coefficients reflect a loss of tightly connected local networks, implying disrupted local specialization. The predominance of negative values suggests widespread local network degradation, particularly in frontal and medial temporal regions, which are heavily implicated in AD pathology.

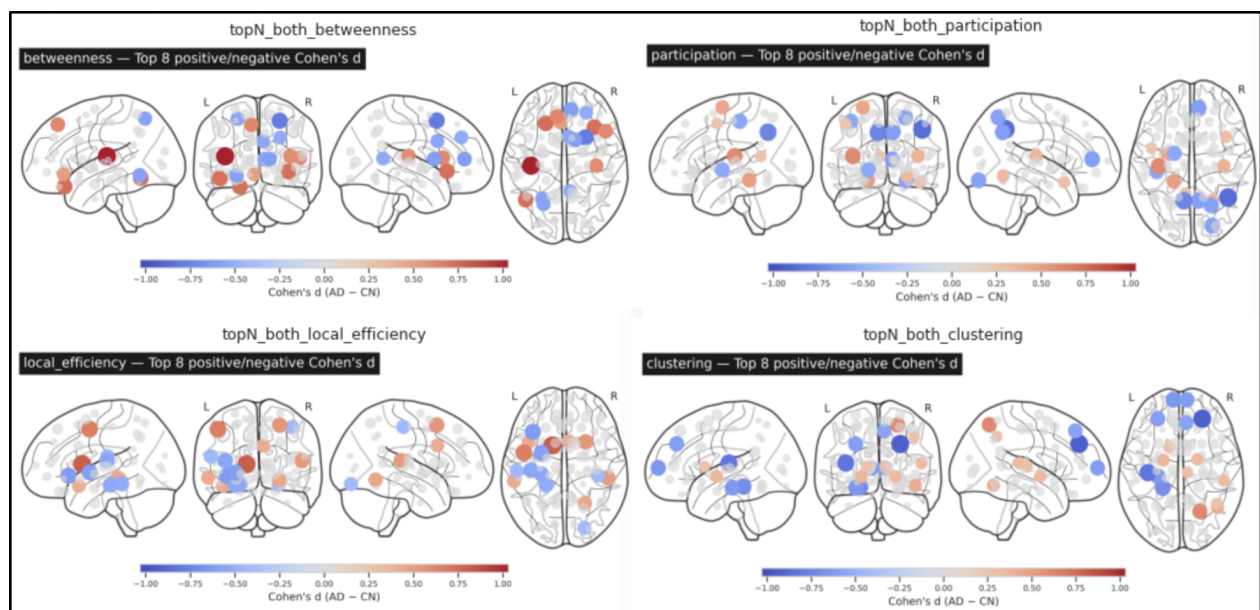


Figure 4b: Functional connectome visualization of group differences. Group differences as measured by Cohen's d are visualized based on their physical location in the brain. Each dot represents a node/brain region. Graph theory measures include betweenness centrality (top left), participation coefficient (top right), local efficiency (bottom left), and local clustering coefficient (bottom right). Here, the regions associated with the top eight most positive and negative standardized mean differences per metric (16 total) are included. Red dots represent a higher AD value, while blue dots represent a higher CN value. Axial, posterior, and lateral views are displayed. Scaling is consistent across all plots.

As shown in Table A1 in the Appendix, there are several nodes of medium group difference in the temporal, sensorimotor, and subcortical sectors of the brain. In the temporal and subcortical sectors of the brain, results indicate higher standardized mean values in both the AD and CN

groups. In contrast, most of the nodes in the sensorimotor sector are associated with higher values in the AD group. The parietal, occipital, and sensorimotor sectors show an overall increased standardized mean difference by a magnitude of 10 or 10^2 in comparison to the other sectors. The parietal and sensorimotor sectors show a standardized mean difference belonging to the AD group while the occipital sector has a standardized mean difference belonging to the CN group.

According to Table A2, the sensorimotor and subcortical sectors have 62.5% of all nodes above a standardized mean difference of above 0.3. Among these, an overwhelming number of nodes in the subcortical sector belong to the CN group. The sensorimotor cortex, however, has 64.3% of the nodes in the CN group. In general, all sectors have more nodes in the CN group than the AD group for the clustering coefficient standardized mean. The parietal and subcortical sectors show an overall increased standardized mean difference of greater than 0.2.

As shown in Table A3, there are no sectors with many nodes with $|d| > 0.3$. The parietal, subcortical, and other sectors with the highest amount only amount to a mean Cohen's d of -0.07. Overall, the number of nodes that are either in the AD or CN group are split evenly for the local efficiency nodes.

As shown in Table A4, none of the sectors have a high number of nodes with a moderate value of Cohen's d ($|d| > 0.3$). Most nodes belong to the CN group apart from the temporal and sensorimotor sectors, although every sector is almost split evenly for which nodes are in the AD or CN group. The mean Cohen's d is very low in magnitude with the sector with the highest magnitude being the subcortical sector with a value of -0.2.

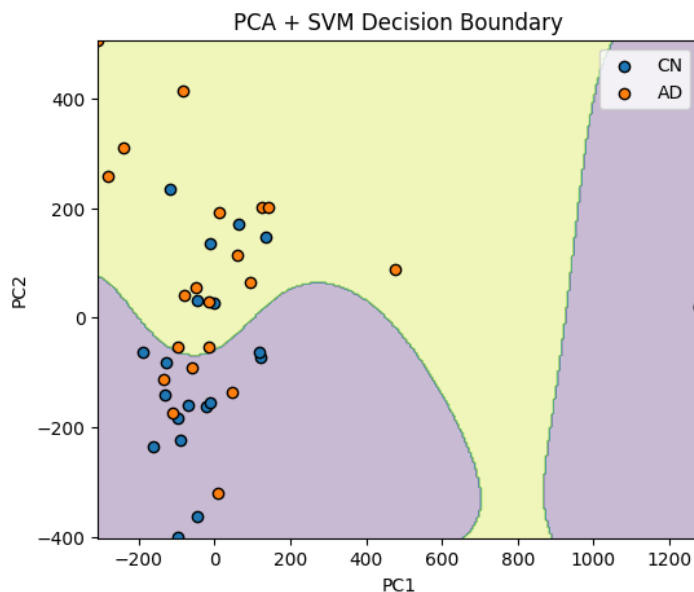


Figure 5: Principal Component Analysis using two principal components. The first principal component PC1, captures the most variance in the data while the second principal component PC2, captures the second most variance in the data. The two classes of control and Alzheimer's subjects are visualized on a nonlinear decision boundary. Subject data is plotted on two regions where the subjects on the green

region represent where the model predicts that subjects have Alzheimer's disease, and the purple region represents where the model predicts that subjects do not have Alzheimer's disease.

The machine learning model achieved an accuracy of 0.75. Figure 5 shows a simplified two-dimensional visualization of the SVM model results. While this visualization does not capture the full model, it does capture the visual representation of how well the model classifies individuals. In this graph it can be observed that many patients are misclassified.

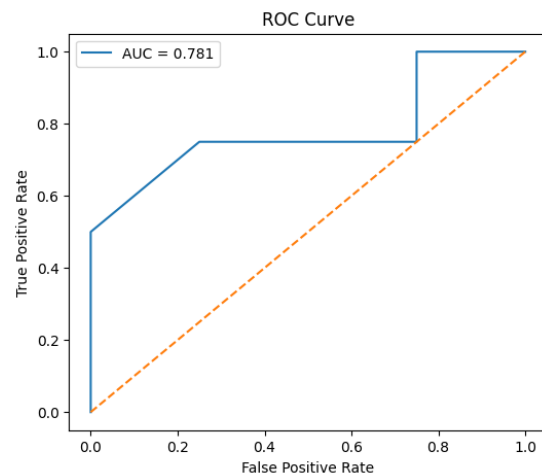


Figure 6: ROC curve of support vector machine model. Model shows a Receiver Operating Characteristic (ROC) curve. Curve shows the trade-off between true positive rate (sensitivity) and false positive rate ($1 - \text{specificity}$). The ROC curve is plotted across the false positive rate. The model achieved an Area Under the Curve (AUC) of 0.781. The dashed orange diagonal represents a random classification (AUC=0.5).

The SVM classifier is further tested using the Receiver Operating Characteristic (ROC) curve. The model achieved an area under the curve (AUC) of 0.781 which indicates moderate ability to differentiate between classes from thresholds of 0 to 1.0 increasing in 0.01 increments. According to Figure 6, the shape of the curve indicates that the model maintains a high amount of true positive rates at low positive rates, suggesting that it can identify a subset of AD subjects with high confidence. However, the curve plateaus across thresholds above 0.3, reflecting limited ability for the model to differentiate from true positives to false positives.

Discussion

The goal of this study was to identify differences in Alzheimer's disease patients and control individuals based on functional neuroimaging data. The main approach used graph theory measures as features for a binary support vector machine classification model. First, the functional connectivity of all subjects in the AD and CN group were calculated and compared. Then, the ROIs with the top standardized group differences were found across a range of graph theory measures and shown on a horizontal bar plot. To further show where groups of ROIs

showed the largest standardized group differences, the regions associated with the top eight most positive and negative standardized mean differences per metric were plotted on a glass brain visualization. Finally, the binary support vector machine classification model was implemented with all analyzed graph theory measures as features. The model was used to classify patients' functional connectivity as belonging to the AD or CN group.

The averaged functional connectivity plots of the 20 subjects in the AD group and 20 subjects in the CN group showed a general modular organization of brain regions. This suggests that clusters of regions are formed that are strongly connected to each other, implying that the brain is organized into modules specialized for similar functions (Betzel et al., 2017). Furthermore, the functional connectivity between ROIs within clusters have higher functional connectivity, which suggests that clusters of ROIs have high segregation. This implies that ROIs within clusters specialize for specific function tasks more efficiently. Finally, weak out-of-cluster functional connectivity between nodes suggests low integration, which implies that communication between modules is more selective. This balance is what allows efficient communication between brain regions (Barabino et al., 2024).

For the clustering coefficient in Figure 4a, the ROIs with the largest group difference are negative which indicates lower local clustering in the AD group relative to controls. Furthermore, if nodes in the AD group have lower local clustering, there is reduced local connectivity and segregation between local ROIs. This may reflect that network degradation is observed, which disrupts local circuits, leading to synaptic loss (Meftah & Gan, 2023). The ROIs with the greatest magnitude of effect sizes are observed in the prefrontal and medial temporal sectors which are responsible for executive function and are known to be impaired in Alzheimer's disease (Flores et al., 2022). These findings support the idea that the functional connectivity of these sectors, which is responsible for cognitive function in AD patients, are associated with a breakdown of local network organization. Furthermore, disruptions of local network organization in these sectors may contribute to impairments in memory and executive function, which are primary symptoms of Alzheimer's disease. In Figure 4b, these findings are further supported. Observing the spatial distribution across the brain, increases in clustering in AD are localized to specific regions. This indicates heterogenous but overall reduced local connectivity in AD.

For local efficiency, Figure 4a shows that the ROIs with the largest group difference are negative, indicating lower local efficiency in the AD group relative to controls. Furthermore, if nodes in the AD group have lower local efficiency, this implies that there is less efficient communication among neighboring ROIs in AD. This suggests that even if connections exist, they are not optimal. The largest magnitude effect sizes are observed in subcortical and frontal sectors in the brain. This supports the idea that the functional connectivity of these sectors, which are responsible for executive functions and motor control, shows local network disruption in AD. Furthermore, these local network disruptions have been associated with impairments in executive processing and may contribute to motor slowing and deficits in learning, which are commonly observed in Alzheimer's disease (Xiong et al., 2022). Results shown in Figure 4b further

supports this result. It is shown that the effect sizes are evenly distributed among red and blue nodes which suggests heterogeneous effects of local efficiency on nodes. Overall, this shows reduced local efficiency in AD.

Considering betweenness centrality results, in Figure 4a, the ROIs with the largest group difference are predominantly positive, although there are a portion of nodes with negative group differences, showing that some brain regions have increased betweenness centrality for the AD group. The largest magnitude effect sizes are observed in subcortical, temporal, and sensorimotor sectors in the brain. Overall, the effect sizes are evenly distributed. These findings support the idea that the functional connectivity of all sectors have relatively consistent impact on the brain.

Finally, observing participation coefficient in Figure 4a, the ROIs with the largest group difference are predominantly negative showing that brain regions in the AD group have decreased participation coefficient overall. The largest magnitude effect sizes are observed in subcortical and temporal sectors. These findings support the idea that the functional connectivity of these sectors, which are responsible for memory processing and information integration, are associated with reduced integration and cross-module connectivity in AD patients. Furthermore, this can cause memory impairment and deficits in cognitive processing which are symptoms of Alzheimer's disease (Jack et al., 2018).

In Figure 5 both CN and AD subject data are plotted against a nonlinear decision boundary in a two principal component feature space. There are many subjects that have been misclassified across both groups. This is consistent with the moderate accuracy of 75% that the SVM model achieved. Misclassification is expected in this visualization as the PCA projection only captures a subset of the total variance of the data (Jolliffe & Cadima, 2016). The subset of variance seen in Figure 5 is further explained in Figure 6. In this figure, the ROC curve plateaus at 0.3, indicating the model struggles to learn a robust decision boundary that divides AD and CN groups across a full range of thresholds. In addition, the model only predicts AD for subjects where there is high confidence, so these predictions are often correct. When the threshold increases, the model is forced to make more uncertain predictions, which results in true positives decreasing and false positives increasing (Saito & Rehmsmeier, 2015).

There are many improvements that can be made to this study. Having minimal subjects results in limited statistical power and risks of overfitting in machine learning. Using a fixed threshold influenced the resulting network topology and statistical significance of the data. Having a range of thresholds would greatly affect graph theory measures as these measures are known to be sensitive to threshold selection (Garrison et al., 2015). Removing negative functional connectivity correlations removes information about negative correlations, which may reflect meaningful brain activity. Furthermore, using static functional connectivity, instead of looking at the changes of functional connectivity over time, overlooks time related dynamics related to brain function. While the SVM model demonstrated moderate predictive accuracy (73%), it is not sufficient for clinical applications. In addition, the PCA visualization only captures a portion of the total variance in the data. Using more than one type of machine learning model would allow more ML approaches to be explored, aiding results.

Future work first and foremost needs to include datasets with many more subjects to improve statistical power. Furthermore, a virtual brain model can be created with longitudinal data with associated graph theory measures to simulate disease progression and predict outcomes for treatments. Using a more robust machine learning model such as XGBOOST may capture more complicated relationships in the data, otherwise not captured with an SVM model. In addition, looking at multimodal data such as using structural MRI, PET, and genetics with rs-fMRI data may improve the interpretability and predictive performance for the machine learning model.

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Appendix

Sector-Level Summary of Cohen's d for Betweenness Centrality (AD vs CN)

Sector	N(d >0.3)	Mean d	% Positive	% Negative
Frontal	3	-0.011172	42.857143	57.142857
Parietal	6	0.23295	71.428571	28.571429
Temporal	7	-0.054099	50	50
Occipital/Visual	4	-0.125195	21.428571	78.571429
Sensorimotor	7	0.143859	71.428571	28.571429
Subcortical	8	-0.00628	46.153846	53.846154
Other	3	-0.039724	53.846154	46.153846

Table A1: Brain sector-level summary of standardized group differences for betweenness centrality in AD versus CN subjects. The column labeled “N(| d |>0.3)” indicates the number of nodes that have a standardized mean difference of moderate value (greater than 0.3). “Mean d ” represents the average standardized group difference of all the nodes in a sector. “% Positive” represents the proportion of nodes in a sector where the AD group displays a higher value. “% negative” represents the proportion of nodes in a sector where the CN group displays a higher value.

Sector-Level Summary of Cohen's d for Local Clustering Coefficient (AD vs CN)

Sector	N(d >0.3)	Mean d	% Positive	% Negative
Frontal	5	-0.108643	35.714286	64.285714
Parietal	6	-0.231394	21.428571	78.571429
Temporal	3	-0.196191	21.428571	78.571429
Occipital/Visual	5	-0.000911	57.142857	42.857143
Sensorimotor	8	-0.175271	35.714286	64.285714
Subcortical	7	-0.254374	15.384615	84.615385
Other	5	-0.164959	23.076923	76.923077

Table A2: Brain sector-level summary of standardized group differences for local clustering coefficient in AD versus CN subjects. “N(| d |>0.3)” indicates the number of nodes that have a standardized mean

difference of moderate value (at least 0.3). “Mean *d*” represents the average standardized group difference of all the nodes in a sector. “% Positive” represents the proportion of nodes in a sector where the AD group displays a higher standardized mean value. “% Negative” represents the proportion of nodes in a sector where the CN group displays a higher standardized mean value.

Sector Level Summary of Cohen’s *d* for Local Efficiency (AD vs CN)

Sector	N(<i>d</i> >0.3)	Mean <i>d</i>	% Positive	% Negative
Frontal	3	0.01429	64.285714	35.714286
Parietal	6	-0.006535	42.857143	57.142857
Temporal	5	-0.067516	42.857143	57.142857
Occipital/Visual	5	0.07799	57.142857	42.857143
Sensorimotor	3	-0.006066	50	50
Subcortical	6	-0.074114	61.538462	38.461538
Other	6	0.012224	46.153846	53.846154

Table A3: Brain sector-level summary of standardized group differences of local efficiency in AD versus CN subjects. “N(|*d*|>0.3)” indicates the number of nodes that have a standardized mean difference of moderate value (at least 0.3). “Mean *d*” represents the average standardized group difference of all the nodes in a sector. “% Positive” represents the proportion of nodes in a sector where the AD group displays a higher standardized mean value. “% negative” represents the proportion of nodes in a sector where the CN group displays a higher standardized mean value.

Sector Level Summary of Cohen’s *d* for Participation Coefficient (AD vs CN)

Sector	N(<i>d</i> >0.3)	Mean <i>d</i>	% Positive	% Negative
Frontal	4	-0.140462	35.714286	64.285714
Parietal	3	0.008991	35.714286	64.285714
Temporal	5	-0.098441	42.857143	57.142857
Occipital/Visual	3	-0.110809	35.714286	64.285714
Sensorimotor	5	-0.09571	50	50
Subcortical	4	-0.200427	15.384615	84.615385
Other	4	-0.076246	30.769231	69.230769

Table A4: Brain sector-level summary of standardized group differences of participation coefficient in AD and CN subjects. “N(|*d*|>0.3)” indicates the number of nodes that have a standardized mean difference of moderate value (at least 0.3). “Mean *d*” represents the average standardized group difference of all the

nodes in a sector. “% Positive” represents the proportion of nodes in a sector **where** the AD group displays a larger standardized mean value “% negative” represents the proportion of nodes in a sector **where** the CN group displays a larger standardized mean value.

Top ROI-Level Effects for betweenness Centrality:

ROI	Cohen's d
RM-A1_L	1.03
RM-PFCdl_R	-0.80
RM-PFCdl_L	0.74
MM82a-G_R	0.73
RM-TCc_L	0.73
RM-PFCdm_L	0.62
RM-PFCm_R	-0.62
RM-PMCvl_R	0.61
BG-Cd_R	-0.61
RM-A1_R	0.59

Table B1: Brain region-level summary of the largest standardized group differences for betweenness centrality in AD versus CN subjects. The table displays the top 10 ROIs ranked by absolute Cohen's *d* value. Positive values indicate that the AD group displays a higher betweenness centrality, while negative values indicate that the CN group displays a higher betweenness centrality.

Top ROI Level Effects for Clustering Coefficient:

ROI	Cohen's d
RM-PFCcl_R	-0.94
RM-A1_L	-0.82
RM-PHC_L	-0.70
RM-HC_L	-0.65
RM-PFCpol_L	-0.65
RM-PCip_R	0.62
RM-PFCpol_R	-0.61
RM-PFCcl_L	-0.61
RM-PFCdm_R	-0.58
RM-PFCdm_L	-0.57

Table B2: Brain region-level summary of the largest standardized group differences for local clustering coefficient in AD versus CN subjects. The table displays the top 10 ROIs ranked by absolute Cohen's *d*

value. Positive values indicate that the AD group displays a higher local clustering coefficient, while negative values indicate that the CN group displays a higher local clustering coefficient.

Top ROI Level Effects for Participation Coefficient:

ROI	Cohen's <i>d</i>
BG-Cd_L	0.81
RM-FEF_L	0.67
MM82a-G_L	-0.62
BG-Pu_L	-0.61
RM-PFCdl_R	0.55
RM-A1_L	-0.55
RM-HC_L	-0.55
RM-A2_R	0.53
RM-PHC_L	-0.52
RM-TCv_R	0.46

Table B3: Brain region-level summary of the largest standardized group differences for participation coefficient in AD versus CN subjects. The table displays the top 10 ROIs ranked by absolute Cohen's *d* value. Positive values indicate that the AD group displays a higher participation coefficient, while negative values indicate that the CN group displays a higher participation coefficient.

Top ROI Level Effects for Local Efficiency

ROI	Cohen's <i>d</i>
BG-Cd_L	0.81
RM-FEF_L	0.67
MM82a-G_L	-0.62
BG-Pu_L	-0.61
RM-PFCdl_R	0.55
RM-A1_L	-0.55
RM-HC_L	-0.55

Table B4: Brain region-level summary of the largest standardized group differences for local efficiency in AD versus CN subjects. The table displays the top seven ROIs ranked by absolute Cohen's *d* value. Positive values indicate that the AD group displays a higher local efficiency, while negative values indicate that the CN group displays a higher local efficiency.